Dataset Profiling Report

EnergyPulse - Smart Energy Consumption Analytics

Project Information

Project Name	EnergyPulse - Smart Energy Consumption Analytics
Module	AID-3 Scalable Chunk Pipeline & Preprocessing
Deliverable	Dataset Profiling & Memory Architecture Assessment
Dataset	UCI Household Electric Power Consumption Dataset
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Version	1.0

1. Objective

Analyze dataset structure, quality, and memory characteristics before building the preprocessing pipeline.

2. Dataset Overview

File Format	TXT (semicolon separated)
File Size	126.80 MB
Total Records	2,075,259
Total Features	9
Sampling Frequency	1 Minute
Data Domain	Residential Energy Consumption
Sample Memory Usage	42.85 MB / 100k rows

3. Key Findings

• Missing values are represented using ? instead of NaN.
• Date and Time were successfully combined into a Datetime field.
• Several numeric columns were interpreted as object data type because of missing-value symbols.
• Chunk-based processing is required for scalability.

4. Engineering Decision

Selected chunk size: 100,000 rows per chunk.

5. Conclusion

Profiling confirmed the need for chunk-based ingestion and established preprocessing requirements for the EnergyPulse pipeline.